

DATA COLLECTION AND SOAP

- Healthcare is "event driven"
 - We go to a physician when something is wrong...
- Healthcare documentation is governed by the **SOAP** principle
 - Subjective
 - Objective
 - Assessment
 - Plan
- Documentation according to "SOAP" is often called "SOAP Note"

- PCP: "what's wrong?"
 - Patient: "I feel ..."
- PCP: "we will do a few measurements ..."
- PCP: "I conclude you have ..." (diagnosis)
- PCP: "Let's do the following ..."

Subjective

Objective

Assessment

Plan

A SOAP Note (paper)

- A SOAP note for an outpatient osteopathic

Outpatient Osteopathic SOAP Note Form

Patient's Name: _____ Sex: ☒ Male ☐ Female For office use only: _____

Date: 11/1/01 Age: 25

Vital Signs: B/P 120/80 Pulse 80 Respir. 20 Temp. 98.6 Wt. 130 Ht. 5'5"

CC: Stomachache, Back Pain @ flexion & extension

HPI: Started 2 days ago, hurts mostly mid epigast. & mid back, & radiation, eating & bed rest help

ROS / PFSH: 10/11 trauma, Dad deceased 46 y.o. of PUD

Meds: Motrin 600mg BID

O: Abd - tender epigastrium @ bowel sounds, & masses. UE bilat muscle strength 5/5

Methods Used to Examine: T A R - passive T - active Severity Scale: 0 No SD or background (BG) levels 1 Obvious TART (esp. R and TL, +/- symptoms) 2 More than BG levels, minor TART 3 Key lesions, symptomatic, R and T stand out

Region	Severity	Somatic Dysfunctions and Other Systems	OMT	Treatment Method	Response
Evaluated	0 1 2 3	MS / SNS / PNS / LYM / CV / RES / GI / FAS / etc	Yes No	(Circle Treatment Methods Used)	R I U W
Head	0			ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
Neck	0	parietal and tender to touch		ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
Thoracic T1-4	0	T12 T13		ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
T5-9	0	T12 T13 (R)		ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
T10-12	0	T10 FSR R		ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
Lumbar	0	myofascial strain		ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
Pelvis/Sacrum	0	WSDL		ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
Pelvis/Innom.	0			ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
Extremity lower	0	Abductor tender TFL		ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
Extremity upper	0	Abduction 160°		ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
Ribs	0			ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	
Other/Abd.	0	↓ mobility stomach greater curvature		ART / BLT / CR / CS / DR / FR / HVLA / IND / INR / LAS / ME / MFR / ST / VLS	

Physician's evaluation of patient prior to treatment: First Visit ☐ Resolved ☐ Improved ☐ Unchanged ☐ Worse ☐

A. 1. Gastritis 3. SD T/R/L/LUE/LC/H/L

P. 2. Thoracic strain 4. _____

OMT performed as above: 1-2 areas ☐ 3-4 areas ☐ 5-6 areas ☐ 7-8 areas ☐ 9-10 areas ☐

Meds: Add Motrin 600mg BID PT: Hot Packs done

Exercise: Shoulder stretches Other: Emotional Support given for 5 minutes

Nutrition: bland diet

Minutes spent with patient: 15 30 45 60 75 90 Follow-up: 1 2 3 4 5 6 8 11 12 Units: Days Wk Mo Yr PPH

Signature of the examiner: SL Sleszynski DO

Joy A. Awoniyi
Ambulatory Care Rotation
Preceptor: Dr. Daryl Norwood

Diabetes Case

Presentation

R.C is a 57-year-old man with Type 2 diabetes first diagnosed two years ago. Other medical problems include obesity and hypothyroidism. He has a history of heavy alcohol use but quit drinking alcohol 2 years ago. He presents now for routine follow-up and is noted to have a blood pressure of 168/100 mmHg. He is asymptomatic.

Physical exam reveals a height of 5'8" (172cm), weight of 243 pounds (110kg), blood pressure of 160/100 mmHg, and a regular pulse of 84 beats per minute. There is no retinopathy or thyromegaly. There is no clinical evidence of congestive heart failure or peripheral vascular disease.

Laboratory evaluation reveals trace protein on urinalysis, a low-density lipoprotein (LDL) cholesterol level of 134 mg/dL, a high-density lipoprotein (HDL) cholesterol level of 35mg/dL, a triglyceride level of 460mg/dL and a total cholesterol level of 240 mg/dL, blood urea nitrogen of 14mg/dL, serum creatinine of 1.2mg/dL, random serum glucose of 192 mg/dL, a glycosylated hemoglobin level of 9.5%, normal electrolytes, and normal thyroid-stimulating hormone levels. A 24-hour urine collection reveals a urinary albumin excretion rate of 250mg daily.

Subjective and Objective Findings

- Patient is asymptomatic
- PMH of alcohol abuse (quit 2 years ago), obesity, and hypothyroidism
- Pertinent vital signs and Lab Values
 - Blood Pressure: 168/100
 - Regular pulse: 84 bpm
 - Serum Creatinine: 1.2mg/dL
 - Creatinine Clearance calculated at 105.6mL/min or 65.7mL/min (Based on ABW and IBW, respectively)
 - BMI was calculated to be 36.94kg/m²
 - Possible relation to hypothyroidism
 - Albumin excretion of 250mg/day (normal levels between 50 and 80mg/24hrs)
 - LDL: 134 mg/dL
- Diagnostic Tests: HgA1C of 9.5%

Assessment

- Related Complications
 - High blood pressure
 - Dyslipidemia
 - Nephropathy manifested as proteinuria
- Risk Factors
 - Obesity (BMI greater than 25)
 - A1C >= 5.7%
 - Blood pressure >= 140/90; Hypertension
- Therapeutic Goals
 - Tight glycemic control: Reduce A1C to less than 7% without causing hypoglycemia
 - Prevention of cardiovascular disease

Diabetes Case SOAP	1
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<https://www.slideshare.net/JAwoniyi/diabetes-soapnote-exersice>

Plan

Further Tests and Work-Up

- Liver function tests
- Repeat blood pressure to confirm hypertension
- CBC to rule out and monitor for infections
- More detailed history needed
 - List of medications (current and failed therapies)
 - Family history
 - Previous immunizations
 - Medication Regimen adherence and barriers to adherence

Treatment Recommendation:

To manage his diabetes, I recommend that the patient be started on **Metformin 500mg once daily** in addition to the initiation of lifestyle modifications. This dose may be increased to twice daily dosing as needed or as tolerated every 1-2 weeks with a maximum dose of 2 grams daily. In the management of diabetes for this patient, drugs that may cause the patient to gain weight should be avoided.

The patient's blood pressure should be managed with an ACE Inhibitor such as **Lisinopril 10mg daily**. Once daily dosing would be ideal and his choice of ACE-inhibitor should be related to the price for the patient as most ACE inhibitors exhibit similar profiles. Hydrochlorothiazide may be added for additional control and numerous preparations are available as combinations of ACE-inhibitors and thiazide diuretics. A beta-blocker is not recommended at this point in therapy due to the masking of hypoglycemia that may accompany a new diabetes regimen.

Drug therapy is also recommended for this patient to control cholesterol since the values are above 130. A reduction of LDL to 99 from 134 is a 26.11% reduction. In a comparison evaluating multiple trials with endpoints to reduce cardiovascular disease risks in patients with diabetes, Simvastatin appears to have showed the most benefit in the 4S-DM study where doses between 20 and 40mg daily demonstrated an absolute risk reduction of 42.5% and LDL reduction of 36%. **Simvastatin 20mg daily** is recommended as initial treatment for this patient.

Although diabetics are prone to increase in clot formation, Aspirin daily is not recommended at this point in therapy for this particular patient. He is not at high risk and has no evidence of CHF or peripheral vascular disease.

Goals and Monitoring Parameters

- Metformin use requires routine monitoring of liver function tests and serum creatinine. The urine should also be monitored for glucose and ketones.
- ACE inhibitor therapy requires monitoring of electrolyte levels (potassium, in particular), blood pressure, renal function, and BUN.
- Simvastatin use warrants the monitoring of LFTs, as well as Creatinine Kinase, which is indicative of myopathies associated with the medication.

Joy A. Awoniyi
Ambulatory Care Rotation
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- Weight changes should be monitored to assess the need for more aggressive treatments or diet restriction.
- Monitor Carbohydrate and fat intake. Total fat should be less than 7% of the total calories.
- Target Levels:
 - A1C → Less than 7% → Monitor every 2-3 months.
 - Blood pressure → Less than 130/80 → Monitor at every routine visit
 - LDL → Less than 100 → Lipid assessments may be repeated every 1-2 years
- Routine tests should be performed to evaluate the efficacy of therapy as well as to monitor the progression of diabetes to prevent further complications
 - Eye exams annually
 - Serum creatinine at least annually
 - Foot examination
 - Screening for neuropathies

Patient Counseling

- To reduce GI side effects, take metformin with food.
- Avoid drinking alcohol. In addition to its contributions to hyperglycemia, alcohol has a negative interaction on the drugs Metformin and Simvastatin that can cause liver toxicities.
- To achieve adequate control of diabetes, lifestyle modifications are an important part of therapy. Dietary restrictions, increase in physical activity, and gradual weight loss will be more beneficial than taking medication, alone.
 - Exercise should include aerobic activities, such as swimming, walking, or running, at least 3 days during each week.

- Describes the patient's current condition (usually in narrative form) as described by the patient (**subjective**)
- This section usually includes the patient's **chief complaint**, or reason(s) why they came to the physician
- Includes:
 - Onset (when started)
 - Chronology (improvement, worsening)
 - Quality and severity (e.g. "pain rating")
 - Modifying factors (drugs, coffee, stress ...)
 - Additional symptoms

- Although the information primarily comes from the patient, the physician can of course ask relevant questions
 - "do you also feel pain in the neck?"



General

Locking History

Date	4/13/2011	Seen By	Richard, Matthew N.	...	
Date of Onset	2/3/2011	Visit Type	Medicare	...	
Claim Number	BCBS-1234	Alerts			

S



An 83-year-old diabetic female presents today stating that she would like diabetic foot care.

- The second step is usually to perform measurements (objective, repeatable, and traceable facts about the patient's status)
 - Vital signs
 - Findings from physical examinations
 - Results from laboratory
 - Measurements, height/weight/BMI of the patient
 - Other information, like age of the patient
 - Current medications and allergies
 - Family history
 - Everything that can be measured/collected and can have an influence

O

On examination, the lateral aspect of her left great toenail is deeply ingrown. Her toenails are thickened opaque. Vibratory sensation appears to be intact. Dorsal pedal pulses are one over four. There is no hair growth seen on her toes, feet or lower legs. Her feet are warm to the touch. All of her toenails are hypertrophic, opaque, elongated and

2 Patton, Christina E. (Christy) F 8/2/2011 (27 months) [M Parker], Office Visit Established (PS) in Exam 4

Timed Set

14:47:35 (M Parker)

New Set

Meds

Motrin

Allergies

Vital Signs/Chief Complaint (PS)

Date Ordered: Today

Page 1/2

Chief complaint: : Sore throat beginning 3 days ago,

Weight : 33Lb 8oz Lb-oz

Height : 29.5 inches

Head circumference: 19.5 inches

Temperature: 98.6 F

Heart rate: 77 bpm

Respiratory rate: 22 rpm

Temporal

Eng

Done

Orthostatic BP/Vitals Labs Any Trmt Other Options

- Usually, EMR systems have different tabs for the "Objective" part, like:
 - Vital signs (blood pressure, temperature, heart rate, ...)
 - Physical examination
 - Often "normal / abnormal" results
 - Laboratory results
 - ECG
- ...



The screenshot shows a software interface for an EMR system. At the top, there is a header bar with a patient photo on the left and a red 'X' and a green checkmark on the right. Below the header, there are four main input fields: 'Add Or Select An Existing Case (If appropriate)' with the text '1st February 2015 - Significant Lameness Right Fore', 'Date' with the value '17/12/2014', 'Gait Check', and 'Lameness'. Below these fields is a row of icons representing different medical tests: a stethoscope, a syringe, a fork and knife, a blood pressure cuff, a bottle, a folder, a flask, a document, a hammer, a tooth, a thermometer, a plant, and a heart. The text 'Vital Signs for Arabella' is visible below the first few icons. To the right of the plant icon is a button labeled 'SOAP'.

- The Physician's **medical diagnoses** for the medical visit on the given date of a note written
- Narrative text and/or coded (when coded, most for reimbursement purposes)
 - ICD-10 (International Classification for Diseases)
- We will later learn about SNOMED-CT annotations

- This describes what the health care provider will do to treat the patient – ordering labs, referrals, ...
- procedures to be performed
- medications prescribed
- Therapies started

- a patient being reviewed following an appendectomy.
 - Components:
 - Diagnostic component: "continue to monitor labs"
 - Therapeutic component: "advance diet"
 - Referrals: "follow up with Cardiology within three days of discharge for stress testing as an out-patient"
 - Patient education component: "progress is good"
 - Disposition component: "discharge to home in the morning"

IMS (FOR, PhysicalTherapy) (Patient: CAFFERY, JENNET)

Action View Setup Activities Billing Reports Utilities Windows Help

Visit Note (Jan 30, 2011 19 of 19) (Performing: JB)

CAFFERY, JENNET Female 40 y(s) 3 mo(s) 100-01-2106 None Pt. Credit: 160.00

General Jan 30, 2011 (Procedure: Manual Therapy Case: GENERAL 02) QReminder NA

SUBJECTIVE

Patient: Caffery, Jenet (15401)
Gender: Female
DOB: Oct 22, 1970 Age: 40 year 3 month
Address: Street 12106, Chandler AZ 85224

Insurance: BC/BS OF ARIZONA
Primary Dr.: Christina WRIGHT

OBJECTIVE

Shoulder:

Shoulder JENNET complains of pain and discomfort located over the right shoulder. Pain radiates towards the right lower neck. The precipitating event was a MVA. It happened a day ago. The vehicle got hit from driver's side. The accident occurred when vehicle lost control. Patient was the driver wearing a seatbelt. The car was totaled. She describes it as aching feeling, sharp pain, shooting pain and throbbing pain. According to patient, symptoms are constant (75-100%). Discomfort is mild-moderate in nature. Symptoms are much worse than before. There is marked impairment. On a scale of 1 to 10, ten being the worst pain imaginable, the patient rated his pain as 8. She feels better with lying supine, rest and warm baths. She feels worse with any activity or movement, bearing weight, bending over, bending to the sides, climbing stairs and extension. Activities hindered are sleeping and walking, and right upper arm.

OBJECTIVE:

Shoulder:

Inspection of the right shoulder joint reveals bruising. Examination of right shoulder Active ROM shows:
Restricted abduction (175/180) with pain.
Restricted adduction (75/75) without pain.
Restricted flexion (175/180) with pain.
Restricted extension (55/60) without pain.
Restricted external rotation (85/90).
On palpation right shoulder reveals tenderness over anterior aspect.
Pain from right shoulder radiates to arm.
Neuro-vascular assessment of the right shoulder reveals diminished axillary sensation.
On right shoulder load and shift for posterior is 2+.

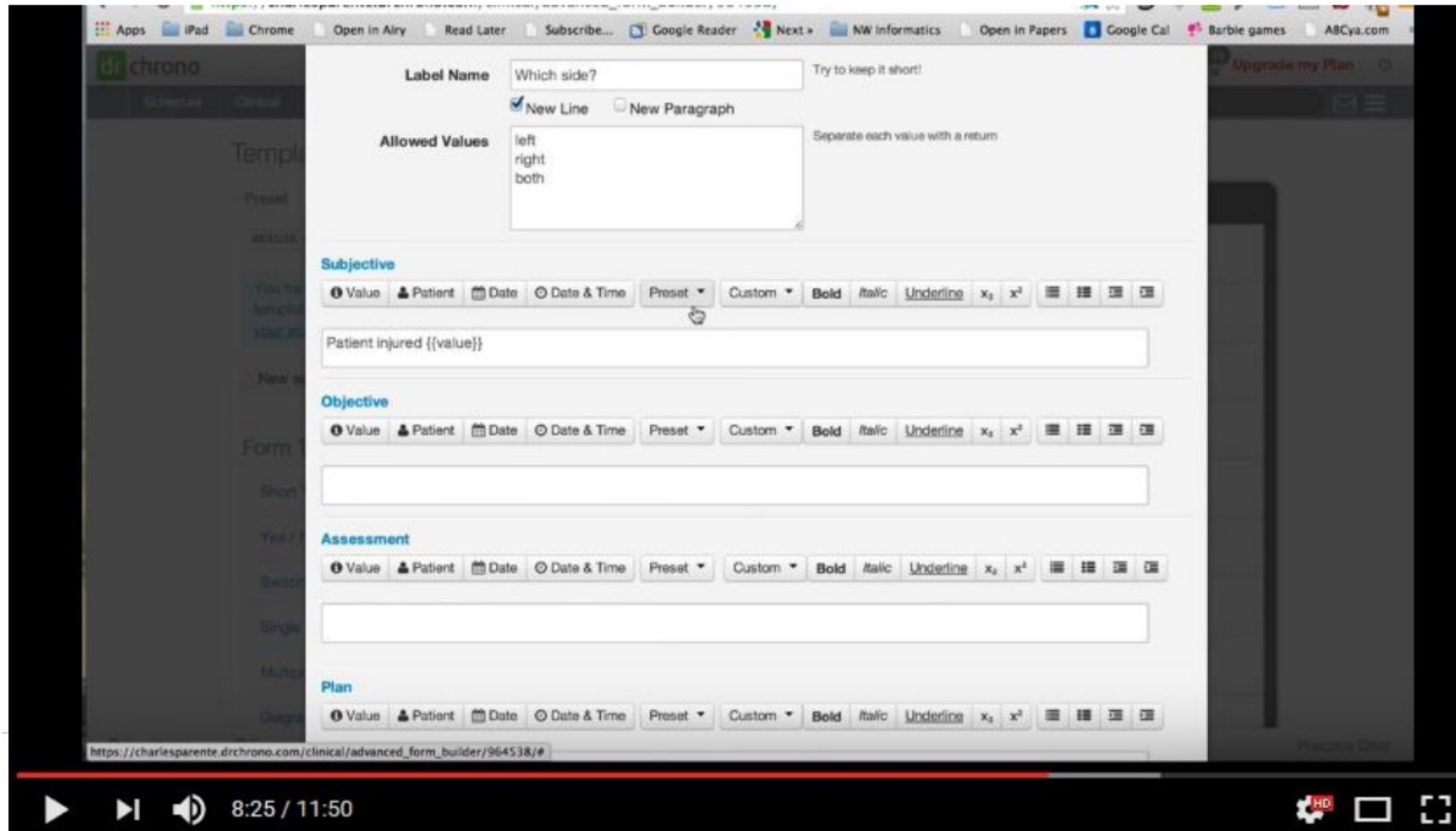
Child Consent to Treat
Sample Form

Ready system Ver: 12.0.5.0331 hp - 9999901 1-30-11 0:49:57
Start IMS (FOR, PhysicalTh... Physical Therapy Check I... Desktop 12:49 AM

- One SOAP note may be the basis of the next SOAP note (subsequent visits)
- **Subjective**
 - status of prior complaints: improvement / worsening
 - New complaints?
- **Objective**
 - Do the **results** show a disease improvement / worsening?
- **Assessment**
 - Was the prior diagnosis correct?
 - Must it be adapted / extended
- **Plan**
 - Must the treatment plan be adapted?

- Some EMR systems allow to build customized SOAP templates
- For example, a department specialized in injuries may want to build an "injury SOAP template".
- Advantages:
 - Less typing, more (automated) coding
 - Standardized questions
 - Completeness
 - Standardized therapies
 - ...

- <https://www.youtube.com/watch?v=6cBVLfvCAZc>



- For reading progress notes, (many) physicians prefer the APSO format (Assessment, Plan, Subjective, Objective)

- Article: "**Physicians rethinking the progress note**"

<http://www.healthcareitnews.com/news/rethinkingprogress-note>

- Article: "**Clinical Leaders Discuss SOAP vs. APSO**"

<http://www.healthcare-informatics.com/article/clinicalleaders-discuss-soap-vs-apso>

Discussion: how would you resolve this issue?

- Writing a SOAP note
- Keep the SOAP note – you will need it later in the course